**Example 1.11**

Add : $\frac{-5}{9}, \frac{-4}{3}, \frac{7}{12}$

***Solution:***

LCM of $9, 3, 12 = 36$

So, $\frac{-5}{9} + \frac{-4}{3} + \frac{6}{12} = \frac{-5}{9} \times \frac{4}{4} + \frac{-4}{3} \times \frac{12}{12} + \frac{7}{12} \times \frac{3}{3}$

$= \frac{-20}{36} + \frac{-48}{36} + \frac{21}{36} = \frac{-20 - 48 + 21}{36} = \frac{-47}{36}$

### 1.3.2 Additive Inverse

The additive inverse of a rational number is another rational number which when added to the given number, gives zero.

For example, $\frac{4}{3}$ and $\frac{-4}{3}$ are additive inverses of each other, since their sum is zero.

**Think**



Is zero a rational number? If so, what is its additive inverse?

### 1.3.3 Subtraction

Subtraction is simply adding the additive inverse.

**Example 1.12**

Subtract : $\frac{9}{17}$ from $\frac{-12}{17}$

***Solution:***

Now, $\frac{-12}{17} - \frac{9}{17} = \frac{-12}{17} + \left(\frac{-9}{17}\right) = \frac{-12 - 9}{17} = \frac{-21}{17}$

**Example 1.13**

Subtract : $\left(-2\frac{6}{11}\right)$ from $\left(-4\frac{5}{22}\right)$

***Solution:***

Now, $\left(-4\frac{5}{22}\right) - \left(-2\frac{6}{11}\right)$

$= \frac{-93}{22} - \left(\frac{-28}{11}\right)$

$= \frac{-93}{22} + \frac{28}{11} = \frac{-93 + 28 \times 2}{22}$

$= \frac{-93 + 56}{22} = \frac{-37}{22} = -1\frac{15}{22}$

### 1.3.4 Multiplication

Product of two or more rational numbers is found by multiplying the corresponding numerators and denominators of the numbers and then writing them in the standard form.



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